POSITIVITY OF THE QUARTIC SOLUTION OF A HIGHER DISCRETE PAINLEVÉ EQUATION

ABSTRACT. We prove positivity of the solution of the quartic discrete Painlevé-type recurrence arising from the quantum minimal surface $w_2 = w_1^3$. The initial data are given by a product of Bessel functions of orders $\pm\frac{1}{4}$. The proof converts the Bessel amplitude into a positive Laplace integral and introduces two auxiliary quartic source variables. The four residue classes modulo 4 then reduce to a single family of bi-moment matrices. A triangular factorization, followed by Andréief's identity, fixes the strict sign of every relevant principal minor. Consequently all signed orthogonal-polynomial norms are positive, and their consecutive ratios give $v_n(a) > 0$ for every $a > 0$ and $n \geq 0$.

1. INTRODUCTION

Discrete Painlevé-type recurrences occur in matrix regularizations of minimal surfaces and in the recurrence coefficients of orthogonal polynomials. For the quantum minimal surface associated with the holomorphic curve $w_2 = w_1^{d-1}$, Felder and Hoppe [5] constructed an indefinite Hermitian form whose orthogonal-polynomial norm ratios obey

$$V_n + \frac{1}{a^2} \sum_{j=0}^{d-2} \prod_{k=0}^{d-2} V_{n+j-k} = \frac{n+1}{a}.$$

This connects the discrete equation with the operator formulation of quantum minimal surfaces originating in [3]; see also [2] for special solutions in the cubic case.

For $d = 4$, after setting $v_n = V_n/a$ and $\epsilon = a^{-2}$, the recurrence becomes

$$v_n \left(1 + v_{n-1}v_{n-2} + v_{n+1}v_{n-1} + v_{n+2}v_{n+1}\right) = (n+1)\epsilon, \qquad n \geq 0, \tag{1.1}$$

where $v_n = 0$ for $n < 0$. The two initial values are determined by

$$v_0 = -\frac{1}{a}\frac{h'(a)}{h(a)}, \qquad v_1 = -\frac{1}{a}\frac{h''(a)}{h'(a)}, \tag{1.2}$$

with

$$h(a) = \frac{\pi^2 a}{4}\Big[J_{-1/4}\left(\frac{a^2}{4}\right)^2 - \sqrt{2}\, J_{-1/4}\left(\frac{a^2}{4}\right) J_{1/4}\left(\frac{a^2}{4}\right) + J_{1/4}\left(\frac{a^2}{4}\right)^2\Big]. \tag{1.3}$$

Felder and Hoppe proved the required Gram nondegeneracy for $d = 1, 2, 3$ and formulated it as a conjecture for general d. The quartic case is the first one not covered by their argument.

Our result is the following.

Theorem 1.1. *For every $a > 0$, the sequence determined by (1.1)–(1.3) satisfies*

$$v_n(a) > 0 \qquad \textit{for every } n \geq 0.$$

2020 *Mathematics Subject Classification.* Primary 34M55; Secondary 33C45, 15B52, 53A10.

Key words and phrases. discrete Painlevé equation, indefinite orthogonal polynomials, quartic potential, total positivity, quantum minimal surface.

The main point is an integral sign mechanism that applies uniformly to all indices. The Bessel expression (1.3) has a positive Laplace representation. Adding independent source variables for z^4 and $\bar{z}^4$ produces four positive integral kernels, one for each residue class modulo 4. Their mixed derivative matrices are reduced by triangular row operations to

$$\left[\int_0^\infty s^p \left(\frac{s^2}{1+s^2}\right)^q \mathrm{d}\mu(s)\right]_{p,q=0}^{k-1}.$$

Both s and $s^2/(1+s^2)$ are strictly increasing. Andréief's identity therefore expresses the determinant as an integral of a product of two Vandermonde determinants with a fixed positive sign. This proves the required sign pattern for every finite block at once.

The paper is organized as follows. Section 2 recalls the quartic Hermitian form and its relation to the recurrence. Section 3 proves the abstract sign lemma. Section 4 derives the four positive kernels and applies the lemma to the four symmetry blocks. Section 5 then identifies the positive norm ratios with the solution of (1.1).

2. The quartic Gram form

Fix $a > 0$ and write

$$\rho_a(z) = \exp\left(-a|z|^2 + \frac{\mathrm{i}}{4}(z^4 + \bar{z}^4)\right). \tag{2.1}$$

The Gaussian factor makes all polynomial moments absolutely convergent. Let $\eta = \mathrm{e}^{\pi\mathrm{i}/4}$. Decompose the polynomial ring into C_4-weight spaces

$$\mathbb{C}[z] = \bigoplus_{r=0}^{3} \mathcal{P}_r, \qquad \mathcal{P}_r = \{f \in \mathbb{C}[z] : f(\mathrm{i}z) = \mathrm{i}^r f(z)\}.$$

For $f \in \mathcal{P}_r$, set

$$(\sigma f)(z) = (-\eta)^{-r} f(\eta z),$$

and define the Hermitian form

$$(f,g)_a = \int_{\mathbb{C}} f(z)\overline{\sigma g(z)}\rho_a(z)\,\mathrm{d}A(z). \tag{2.2}$$

The density is invariant under $z \mapsto \mathrm{i}z$, so distinct weight spaces are orthogonal. If $m = r + 4q$, then

$$\sigma z^m = (-\eta)^{-r}\eta^m z^m = (-1)^{r+q} z^m. \tag{2.3}$$

Thus the Gram matrix on $\mathcal{P}_r$, in the ordered monomial basis $z^r, z^{r+4}, \ldots$, has entries

$$G_{pq}^{(r)} = (-1)^{r+q}\int_{\mathbb{C}} z^{r+4p}\bar{z}^{r+4q}\rho_a(z)\,\mathrm{d}A(z), \qquad p,q \ge 0. \tag{2.4}$$

Assume temporarily that the leading principal minors in all four blocks are nonzero. Gram–Schmidt then gives monic polynomials $P_n(z) = z^n + \cdots$, and symmetry implies $P_n \in \mathcal{P}_{n \bmod 4}$. Define

$$\tau(\ell) = \begin{cases} -1, & \ell \equiv 3 \pmod 4, \\ 1, & \ell \not\equiv 3 \pmod 4, \end{cases} \qquad \varrho(n) = (-1)^n \prod_{\ell=0}^{n-1} \tau(\ell) = (-1)^{n+\lfloor n/4\rfloor}. \tag{2.5}$$

The signed norms h_n are fixed by

$$(P_n, P_n)_a = \varrho(n) h_n. \tag{2.6}$$

Section 4 will prove simultaneously that all the required minors are nonzero and that $h_n > 0$.

3. A sign-regularity lemma

We isolate the integral argument that drives the proof. Let $y > 0$, $\alpha \geq 0$, and $\beta \geq 0$, and define

$$\Phi_{\alpha,\beta}(w;y) = \int_0^\infty \mathrm{e}^{-ys} s^{\alpha-1/2}(1+ws^2)^{-\beta-1/2}\,\mathrm{d}s, \qquad w > 0. \tag{3.1}$$

For $p, q \geq 0$, put

$$D_{pq}^{\alpha,\beta} = \partial_u^p \partial_v^q \Phi_{\alpha,\beta}(uv;y)\big|_{u=v=1}. \tag{3.2}$$

Lemma 3.1 (strict sign regularity)**.** *For every $k \geq 1$,*

$$\operatorname{sgn}\det\left[D_{pq}^{\alpha,\beta}\right]_{p,q=0}^{k-1} = (-1)^{k(k-1)/2}. \tag{3.3}$$

Proof. Set

$$b = \beta + \frac{1}{2}, \qquad \gamma = \frac{\alpha}{2} + \frac{1}{4}, \qquad L(y) = \Phi_{\alpha,\beta}(1;y).$$

The substitution $t = \sqrt{w}\,s$ in (3.1) gives

$$\Phi_{\alpha,\beta}(w;y) = w^{-\gamma} L\left(\frac{y}{\sqrt{w}}\right). \tag{3.4}$$

Let $\vartheta = w\partial_w$, and use the falling-factorial notation

$$(\vartheta)^{\underline{p}} = \vartheta(\vartheta-1)\cdots(\vartheta-p+1).$$

Under (3.4), ϑ acts on L as

$$\mathcal{A} = -\gamma - \frac{y}{2}\partial_y.$$

Since $w = uv$,

$$\partial_u^p \partial_v^q F(uv)\big|_{u=v=1} = (\vartheta)^{\underline{p}}(\vartheta)^{\underline{q}} F(w)\big|_{w=1}.$$

It follows that

$$D_{pq}^{\alpha,\beta} = (\mathcal{A})^{\underline{p}}(\mathcal{A})^{\underline{q}} L(y). \tag{3.5}$$

Define

$$f(s) = \frac{s^2}{1+s^2}$$

and the bi-moments

$$B_{pq} = \int_0^\infty s^p f(s)^q \mathrm{e}^{-ys} s^{\alpha-1/2}(1+s^2)^{-\beta-1/2}\,\mathrm{d}s. \tag{3.6}$$

Direct differentiation of (3.1) with respect to w gives

$$(\mathcal{A})^{\underline{q}} L(y) = (-1)^q (b)_q B_{0q}, \tag{3.7}$$

where $(b)_q$ is the rising Pochhammer symbol. On the other hand,

$$(\mathcal{A})^{\underline{p}} = \sum_{i=0}^{p} c_{pi}(y)\partial_y^i, \qquad c_{pp}(y) = \left(-\frac{y}{2}\right)^p. \tag{3.8}$$

Since $\partial_y^i B_{0q} = (-1)^i B_{iq}$, (3.5)–(3.8) give a factorization

$$D = RBC, \tag{3.9}$$

where R is lower triangular with diagonal

$$R_{pp} = \left(\frac{y}{2}\right)^p > 0,$$

and C is diagonal with

$$C_{qq} = (-1)^q (b)_q.$$

It remains to determine the sign of $\det B$. Andréief's integration identity [1, 6] gives

$$\begin{aligned}\det[B_{pq}]_{p,q=0}^{k-1} = \frac{1}{k!}\int_{(0,\infty)^k} \det[s_j^p]_{p=0,\dots,k-1}^{j=1,\dots,k} \\ \times \det[f(s_j)^q]_{q=0,\dots,k-1}^{j=1,\dots,k} \prod_{j=1}^{k} \mathrm{d}\mu(s_j),\end{aligned} \tag{3.10}$$

where

$$\mathrm{d}\mu(s) = \mathrm{e}^{-ys} s^{\alpha-1/2}(1+s^2)^{-\beta-1/2}\,\mathrm{d}s.$$

The two determinants in (3.10) are Vandermonde determinants. Because both s and $f(s) = s^2/(1+s^2)$ are strictly increasing on $(0,\infty)$, their product is nonnegative and is strictly positive whenever the s_j's are distinct. Hence $\det B > 0$.

The determinant of R is positive, whereas

$$\operatorname{sgn}\det C = (-1)^{0+1+\cdots+(k-1)} = (-1)^{k(k-1)/2}.$$

The factorization (3.9) proves (3.3). □

4. The four positive kernels

We first turn the Bessel initial function into a positive integral. For $x > 0$, the connection formula

$$J_{-1/4}(x) = \frac{J_{1/4}(x) - Y_{1/4}(x)}{\sqrt{2}}$$

reduces (1.3) to

$$h(a) = \frac{\pi^2 a}{8}\left[J_{1/4}\left(\frac{a^2}{4}\right)^2 + Y_{1/4}\left(\frac{a^2}{4}\right)^2\right]. \tag{4.1}$$

The Dixon–Ferrar formula [4, 7] states that, for $|\operatorname{Re}\nu| < \frac{1}{2}$,

$$J_\nu(x)^2 + Y_\nu(x)^2 = \frac{8\cos(\pi\nu)}{\pi^2}\int_0^\infty K_{2\nu}(2x\sinh t)\,\mathrm{d}t. \tag{4.2}$$

Using

$$K_{1/2}(u) = \sqrt{\frac{\pi}{2u}}\mathrm{e}^{-u}$$

in (4.1), and then setting $s = \sinh t$, gives

$$h(a) = \sqrt{\frac{\pi}{2}}\int_0^\infty \frac{\mathrm{e}^{-a^2 s/2}}{\sqrt{s(1+s^2)}}\,\mathrm{d}s. \tag{4.3}$$

Set

$$y = \frac{a^2}{2}, \qquad C = \sqrt{\frac{\pi}{2}}.$$

We now introduce an angular source. For $u, v > 0$, define

$$\mathcal{Z}(a;u,v) = \pi\int_0^\infty \mathrm{e}^{-at} J_0\left(\frac{t^2\sqrt{uv}}{2}\right)\mathrm{d}t. \tag{4.4}$$

This is the angularly integrated two-source expression. Indeed, termwise angular integration gives

$$\int_0^{2\pi} \exp\left(\frac{\mathrm{i}R^4}{4}(u\mathrm{e}^{4\mathrm{i}\theta} + v\mathrm{e}^{-4\mathrm{i}\theta})\right) \mathrm{d}\theta = 2\pi J_0\left(\frac{R^4\sqrt{uv}}{2}\right), \tag{4.5}$$

and $t = R^2$ gives (4.4). Only derivatives at $u = v = 1$ will be used, where all resulting planar moment integrals are absolutely convergent.

At $u = v = 1$, $\mathcal{Z}(a; 1, 1) = h(a)$. Scaling t in (4.4) and using (4.3) yield

$$\mathcal{Z}(a; u, v) = C\int_0^\infty \frac{\mathrm{e}^{-ys}}{\sqrt{s(1 + uvs^2)}}\,\mathrm{d}s = C\Phi_{0,0}(uv; y). \tag{4.6}$$

For $r = 0, 1, 2, 3$, put

$$\mathcal{Z}_r = (-\partial_a)^r \mathcal{Z}.$$

Lemma 4.1. *The four radial source functions are*

$$\begin{array}{c|c} r & \mathcal{Z}_r(a; u, v) \\ \hline 0 & C\Phi_{0,0}(uv; y), \\ 1 & aC\Phi_{1,0}(uv; y), \\ 2 & 2C\Phi_{1,1}(uv; y), \\ 3 & 2aC\Phi_{2,1}(uv; y). \end{array} \tag{4.7}$$

Proof. The first two rows follow by differentiating (4.6). For the third row, integration by parts gives

$$\begin{aligned} y\int_0^\infty \frac{\mathrm{e}^{-ys}s^{3/2}}{\sqrt{1 + uvs^2}}\,\mathrm{d}s = \frac{1}{2}\int_0^\infty \frac{\mathrm{e}^{-ys}s^{1/2}}{\sqrt{1 + uvs^2}}\,\mathrm{d}s \\ + \int_0^\infty \frac{\mathrm{e}^{-ys}s^{1/2}}{(1 + uvs^2)^{3/2}}\,\mathrm{d}s. \end{aligned} \tag{4.8}$$

The boundary terms vanish at 0 and ∞. Since

$$(-\partial_a)^2 = \partial_y + 2y\partial_y^2,$$

equation (4.8) gives $\mathcal{Z}_2 = 2C\Phi_{1,1}$. One more differentiation gives $\mathcal{Z}_3 = 2aC\Phi_{2,1}$. □

For $k \geq 1$, set

$$\Delta_k^{(r)} = \det[G_{pq}^{(r)}]_{p,q=0}^{k-1}, \qquad \Delta_0^{(r)} = 1. \tag{4.9}$$

Proposition 4.2. *For every $r \in \{0, 1, 2, 3\}$ and $k \geq 1$,*

$$\operatorname{sgn}\Delta_k^{(r)} = (-1)^{rk + k(k-1)/2}. \tag{4.10}$$

Proof. Let

$$D_{pq}^{(r)} = \partial_u^p \partial_v^q \mathcal{Z}_r(a; u, v)\big|_{u=v=1}.$$

Every scalar multiplier in (4.7) is positive. Lemma 3.1, applied to the four parameter pairs

$$(\alpha, \beta) = (0, 0), (1, 0), (1, 1), (2, 1),$$

therefore gives

$$\operatorname{sgn}\det[D_{pq}^{(r)}]_{p,q=0}^{k-1} = (-1)^{k(k-1)/2}. \tag{4.11}$$

Differentiating (4.5) at $u = v = 1$ shows that

$$D_{pq}^{(r)} = \left(\frac{\mathrm{i}}{4}\right)^{p+q} \int_{\mathbb{C}} z^{r+4p} \bar{z}^{r+4q} \rho_a(z)\, \mathrm{d}A(z).$$

Combining this with (2.4) gives

$$G_{pq}^{(r)} = (-1)^{r+q} \left(\frac{4}{\mathrm{i}}\right)^{p+q} D_{pq}^{(r)}. \tag{4.12}$$

Let $S_k = k(k-1)/2$. Factoring the row and column phases in (4.12) yields

$$\Delta_k^{(r)} = (-1)^{rk} 16^{S_k} \det[D_{pq}^{(r)}]_{p,q=0}^{k-1}. \tag{4.13}$$

Now (4.10) follows from (4.11). □

Corollary 4.3. *All quartic Gram matrices are nondegenerate, and the signed norms in* (2.6) *satisfy*

$$h_n > 0 \qquad (n \geq 0).$$

Proof. The block sign formula shows first that every $\Delta_k^{(r)}$ is nonzero. Thus all the monic orthogonal polynomials exist. If $n = r + 4j$, Gram–Schmidt within $\mathcal{P}_r$ gives

$$(P_n, P_n)_a = \frac{\Delta_{j+1}^{(r)}}{\Delta_j^{(r)}}. \tag{4.14}$$

Taking the quotient of the signs in (4.10) gives

$$\operatorname{sgn}(P_n, P_n)_a = (-1)^{r+j}.$$

But (2.5) gives $\varrho(r + 4j) = (-1)^{r+j}$. Hence $h_n = \varrho(n)(P_n, P_n)_a > 0$. □

5. Identification with the quartic recurrence

It remains to verify that the positive norm ratios give precisely the sequence in Theorem 1.1. Define

$$V_n = \frac{h_{n+1}}{h_n}, \qquad v_n = \frac{V_n}{a}. \tag{5.1}$$

Corollary 4.3 already gives

$$v_n > 0 \qquad (n \geq 0). \tag{5.2}$$

For clarity, we include the short recurrence calculation from the quartic Gram form. Differentiating (2.1) and integrating by parts in (2.2) give

$$\left((\partial_z + \mathrm{i}z^3)f, g\right)_a = -\tau(\ell) a (f, zg)_a \tag{5.3}$$

when g has C_4-weight ℓ. The Gaussian decay removes the boundary term. Weight, degree, and orthogonality imply

$$\begin{aligned} zP_n &= P_{n+1} + \frac{\mathrm{i}}{a}\frac{h_n}{h_{n-3}} P_{n-3}, \\ (\partial_z + \mathrm{i}z^3)P_n &= \mathrm{i}P_{n+3} + a\frac{h_n}{h_{n-1}} P_{n-1}, \end{aligned} \tag{5.4}$$

where terms with negative indices are omitted. To see the coefficients, pair the first expansion with P_{n-3}, the second with P_{n-1}, and use

$$\varrho(n) = \tau(n)\varrho(n-3), \qquad \varrho(n) = -\tau(n-1)\varrho(n-1).$$

The coefficient of P_{n-1} in z^3P_n receives three contributions: the downward step in the first formula of (5.4) may occur after zero, one, or two upward steps. Hence that coefficient is

$$\frac{\mathrm{i}}{a}V_{n-1}\left(V_{n-2}V_{n-3}+V_nV_{n-2}+V_{n+1}V_n\right).$$

Subtracting $\mathrm{i}z^3P_n$ from the second formula in (5.4), and using that the coefficient of P_{n-1} in $\partial_z P_n$ is n, gives

$$V_m\left[1+\frac{1}{a^2}\left(V_{m-1}V_{m-2}+V_{m+1}V_{m-1}+V_{m+2}V_{m+1}\right)\right]=\frac{m+1}{a}. \tag{5.5}$$

With $V_m = av_m$, this is (1.1).

Finally, $P_0 = 1$, $P_1 = z$, and $P_2 = z^2$. Differentiating $h(a) = (1,1)_a$ gives

$$h_0 = h, \qquad h_1 = -h', \qquad h_2 = h''.$$

Therefore (5.1) gives exactly (1.2). Conversely, the forward form of (1.1) determines v_{m+2} from the preceding terms whenever $v_m v_{m+1} \neq 0$. The positive sequence (5.1) has this property, so induction identifies it with the sequence determined by (1.2). Together with (5.2), this proves Theorem 1.1.

Remark 5.1. The argument proves only the positivity assertion required in Theorem 1.1. It does not use or claim a classification of all positive solutions, a closed evaluation of the Gram determinants, or asymptotic formulas for v_n.

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